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DIARRHEAL DISORDERS IN CHILDREN

Diarrheal disorders are major contributors of under-5 mortality.

Types :

	Duration	Etiology
Acute	< 7 days	Infections
Persistent	> 14 days	Prolonged infection
Chronic	> 14 days (upto months)	Non-infectious causes (especially malabsorption)

Acute Diarrhea

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- most common cause : **Rotavirus**.
- Other viral causes : Norwalk, adeno, astro and calicivirus.
- most common bacteria : **ETEC** (Enterotoxigenic E coli).

Dysentery : Diarrhea + blood in stool.

- most common cause : *Shigella flexneri*.

DEHYDRATION :

- Primary concern in diarrhea.
- Look, feel, decide & treat.

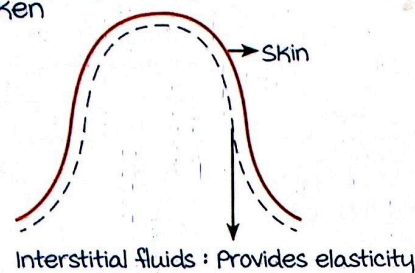
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Assessment :

	No dehydration	Some dehydration	Severe dehydration
Condition	Active & alert	Irritable	Lethargy (Cannot be aroused)
Eyes	Normal	Sunken	Very sunken
Tears	Normal	Absent	Absent
Mouth & tongue	Normal	Dry	Very dry
Thirst	Normal	Severe thirst	Unable to drink
Skin pinch test	Fast (< 1 sec)	Slow (≤ 2 sec)	Very slow (> 2 sec)

Skin pinch testing :

- Pinching the skin → Let go → Evaluate time taken for skin to go back to normal.
- Testing for skin turgor/elasticity.
- Skin turgor depends on interstitial fluid.
- In dehydration : ↓ fluid level → ↓ turgor.



MANAGEMENT :**In no dehydration Plan A :**

- Use **Low osmolarity ORS**.

Composition	Osmolarity
Glucose	75
Na	75
K	20
Cl	65
Citrate	10
Total	245 mosm/L

1:1 ratio for maximum absorption via SGLT-1

- Alternatives of ORS at home :

- Salted rice water.
- Salted yogurt.
- Soup with salt.
- Fresh fruit juice (No sugar).
- Tender coconut water.

Note : **Avoid packed juices** (D/t added sugar).

Amount of ORS :

Principle : Replace ongoing losses.

Age	Amount of ORS/stool
< 2 yrs	50 - 100 mL
2-10 yrs	100 - 200 mL
> 10 yrs	Ad lib

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In some dehydration Plan B :

ORS is used to correct dehydration.

Steps :

1. Rehydration : 75 mL/kg ORS over 4 hrs.
2. Replacement of ongoing loss : Same as plan A.
3. Daily fluid requirements (maintenance) : **Holliday - Segar formula**.

Weight	Daily fluid requirement
First 10 kg	100 ml/kg
10-20 kg	1000 ml (First 10 kg) + 50 ml for every kg after 10 kg
> 20 kg	1500 ml (First 20 kg) + 20 ml for every kg after 20 kg

In severe dehydration Plan C :

1. IV fluids : RL + 5 % dextrose.

- Rehydration : 100 mL/kg for replacement.

	30 ml/kg over	70 ml/kg over	Total time
< 1 year	1 hour	5 hours	6 hrs
> 1 year	30 min	2.5 hours	3 hrs

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2. Zinc : 14 days.

- Plays a role in GI re-epithelialisation.
- ↓ risk of persistent diarrhea.

Age	Dose/day
< 6 months	10 mg
> 6 months	20 mg

Additional drugs :

1. Dysentery : 3rd generation cephalosporin is added.
2. Vomiting : Antiemetics.
 - Ondansetron : 0.15 mg/kg/dose orally.
3. Probiotics : Prescribed based on cost benefit analysis.
 - Lactobacillus rhamnosus & saccharomyces boulardii.

CONVULSIONS WITH DIARRHEA :**Etiology :**

1. Electrolyte and metabolic imbalance :

- Hyponatremia.
- Hypernatremia.
- Hypoglycemia.

2. Infections :

- Febrile seizures.
- meningitis/encephalitis.

3. Dehydration → Cerebral venous thrombosis, sagittal sinus thrombosis.

Persistent Diarrhea

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RISK FACTORS :

1. malnutrition : Causes ↓ immunity.
2. Immunodeficiency.

MANAGEMENT :

1. Fluids : ORS or IV fluids depending upon degree of dehydration.
2. Zinc.
3. Vitamin A :

Weight	Single dose
< 8 Kg	1 lakh IU
> 8 Kg	2 lakh IU

4. Restriction on lactose :

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Persistent diarrhea

↓
Damage to villi↓
lactase

Lactose intolerance → unabsorbed lactose

Colonic bacteria

- H_a formation → Gaseous stool
- Lactic acid in stool
 - ↓ pH.
 - Perianal excoriation.

DIET :

- ↓ lactose : Restriction of milk (< 50 ml/kg/day).

↓
No response

- No lactose diet.
- ↓ sugars and starch.
- Plain glucose, cereals, egg white, chicken.

↓
No response

- No lactose, no starch.
- Only egg yolk/chicken (Protein).

Celiac Disease

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Gluten hypersensitivity causing chronic diarrhea.

Gluten containing foods :

mnemonic : BROW.

- Barley.
- Rye.
- Oats → Relatively well tolerated.
- Wheat → m/c implicated.

Genetic predisposition :

HLA DQ2 and DQ8.

Pathology :

T cell mediated auto-immune disorder.

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Genetic predisposition + gluten consumption → Gliadin $\xrightarrow{\text{Intestinal absorption}}$ T cell stimulation
 → B cell stimulation → Autoantibodies → Intestinal injury.

Associations :

- Down's syndrome.
- Hashimoto's thyroiditis.
- TI DM.
- Turner's syndrome.
- Selective IgA deficiency.

Clinical features :

- Onset $\geq$ 6 months (After stoppage of exclusive breastfeeding).
- malabsorption $\left\{ \begin{array}{l} \rightarrow \text{Failure to thrive} \\ \rightarrow \text{Chronic diarrhea} \end{array} \right.$
- Anemia : D/t iron (Duodenal absorption) and folic acid (Terminal ileum) deficiency.
 - Dimorphic anemia.
 - Can present as **Iron refractory anemia**.
- Failure to thrive (FTT).

Diagnosis :**1. Serology :**

- Screening (Qualitative) : **Anti - TTG** (Tissue transglutaminase) antibody.
- Best : **Anti endomysial Ab.**

Note : Quantitative serology tests (Titres) are seldom preferred.

2. Small intestinal (Duodena) biopsy : Investigation of choice in India.

- villous atrophy (Flattening).
- $\uparrow$ length of crypt.
- $\uparrow$ intraepithelial lymphocytes (> 30 per 100 enterocytes).

Treatment :

Gluten free diet : Lifelong.

Avoid wheat → Substitute with rice & maize.